

Console Tester backlog

This file records work considered after Public Beta 0.7. These items are not part of the current source-cleanup pass and must not silently change the hardware-validated Beta 0.7 program image.

Investigate one missing visible sprite on TI-99/4

Status: Open; observation needs isolation by VDP implementation.

Observed: On the tested TI-99/4, one of the three intended visible sprite groups did not appear during the TMS9918-family sprite exercise. That console currently contains a Pico9918, so this is not yet evidence of native TMS9918 behavior.

The exercise creates five same-scanline sprite entries. Sprites 0 and 1 overlap to stimulate collision, sprites 2 and 3 are separate, and sprite 4 stimulates the fifth-sprite condition. Suppression of sprite 4 is expected on an original VDP; only two visible groups, rather than the intended three, is the unresolved observation.

Investigation:

1. Use the `VDSPRT` reproduction contract in `src/HEXEXT.a99`: SAT `>0300`, pattern table `>1800`, five entries at Y=79, and R1 stages `>E0`, `>E2`, and `>E3`.
2. Reproduce separately on a genuine TMS9918, a Pico9918, a TMS9918A, and an F18A when available.
3. Distinguish sprite-limit behavior from overlap, early-clock, coordinate, pattern, and replacement-VDP compatibility effects.
4. Confirm that the /4 still skips Graphics-II bitmap tests.

Acceptance criteria:

- The intended three visible groups are present where the VDP supports the original behavior, or the screen clearly documents an implementation-specific difference.
- Collision and fifth-sprite status remain observations, not misleading pass/fail claims.
- No stale sprite or table data corrupts the following page.
- The /4, /4A, QI/V2.2, and GROMless recovery paths remain functional.

Revisit Alpha Lock and joystick policy on QI consoles

Status: Open; current Beta 0.7 behavior remains frozen.

Observed: The current page asks the operator to release Alpha Lock and blocks joystick testing while it is down. That is appropriate protection for the original /4A wiring, where Alpha Lock interferes with joystick Up, but users report that the QI motherboard corrects that interference. The same restriction is therefore unnecessarily severe on a QI console.

Required profiles:

Console profile	Question to resolve
Stock TI-99/4	Preserve verified joystick selectors 5 and 6 and appropriate /4 keyboard messaging

Console profile	Question to resolve
Stock TI-99/4A	Prevent Alpha Lock from being mistaken for joystick Up and explain why release is requested
TI-99/4QI / V2.2	Permit useful joystick testing with Alpha Lock down if hardware validation confirms isolation
GROMless, custom, or unknown /4A-family console	Remain useful without assuming firmware identity; prefer an accurate caution/raw display over a false hard block

Firmware checksums may identify a stock V2.2 set, but firmware identity alone may not prove motherboard wiring on a modified console. The investigation must separate what is detected from ROM/GROM contents, what can be probed electrically through the TMS9901, and what must be presented as operator guidance.

Acceptance criteria:

- Joystick Up never creates a false `RELEASE ALPHA LOCK` warning.
- A confirmed QI can exercise both joysticks with Alpha Lock down if physical tests confirm that this is safe and unambiguous.
- An original /4A still receives clear guidance when Alpha Lock masks a valid joystick direction.
- Unknown/GROMless systems retain direct keyboard and joystick visibility and are not made unusable by an uncertain model classification.
- Messaging states the detected profile and the limitation in plain language, without asking users to understand the TMS9901 circuit.
- Holding 0 continues to return to the menu, and no FCTN/CTRL combination is dispatched as a console action.